B. Longest Divisors Interval

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 900 | Tags: brute force, combinatorics, greedy, math, number theory | https://codeforces.com/problemset/problem/1855/B

Given a positive integer n, find the maximum size of an interval $[l, r]$ of positive integers such that, for every i in the interval (i.e., $l \le i \le r$), n is a multiple of i.

Given two integers $l \le r$, the size of the interval $[l, r]$ is $r-l+1$ (i.e., it coincides with the number of integers belonging to the interval).

Input

The first line contains a single integer t ($1 \le t \le 10^4$) — the number of test cases.

The only line of the description of each test case contains one integer n ($1 \le n \le 10^{18}$).

Output

For each test case, print a single integer: the maximum size of a valid interval.

Example

Input
10 1 40 990990 4204474560 169958913706572972 365988220345828080 387701719537826430 620196883578129853 864802341280805662 1000000000000000000

Output
1 2 3 6 4 22 3 1 2 2

Note

In the first test case, a valid interval with maximum size is [1, 1] (it's valid because $n = 1$ is a multiple of 1) and its size is 1.

In the second test case, a valid interval with maximum size is [4, 5] (it's valid because $n = 40$ is a multiple of 4 and 5) and its size is 2.

In the third test case, a valid interval with maximum size is [9, 11].

In the fourth test case, a valid interval with maximum size is [8, 13].

In the seventh test case, a valid interval with maximum size is [327869, 327871].